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A Data Statistics

We provide source information of our data in table 6 and statistic information of data length in 10.

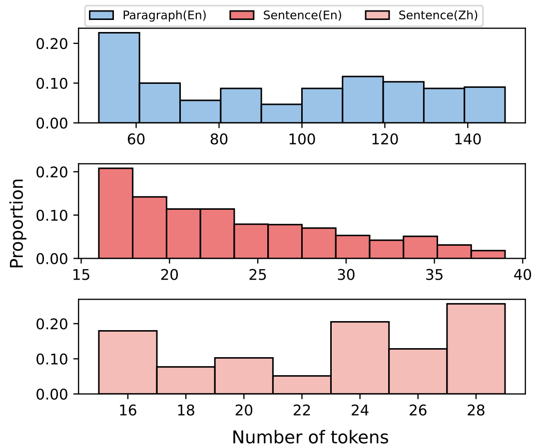

Figure 10: Statistical patterns of data length distribution.

B Change in similarity

We measure the change in similarity between T_i and T_0 across successive paraphrasing steps. The results are presented in Figure 11. As the number of paraphrasing steps increases, most LLMs maintain the similarity between paraphrases and their corresponding original texts, with the exception of an initial drop in similarity. Meanwhile, it also exhibits aslight 2-periodicity in similarity. By combining Figure 11 and Table 2, we found that models with higher periodicity also exhibit higher similarity.

C Generalization

C.1 Task Extentions

We propose four additional tasks beyond paraphrasing: polishing (**Pol.**), clarification (**Clar.**), informal-to-formal style transfer (**I/F.**), and forward/backward translation (**Trans.**). The detailed prompts for these tasks are listed in Table 7. We

Dataset	TLDR	SQuAD	ROCT	Yelp	ELI5	Sci_Gen
Sentence/Paragraph	100/30	100/30	100/30	100/30	100/30	100/30
Dataset	**XSum**	**CMV**	**HSWAG**	**WP**	**Wiki**	**WMT**
Sentence/Paragraph	100/30	100/30	100/30	100/30	200/0	200/0

Table 6: Dataset Setup: Datasets marked in red indicate Chinese datasets, while others represent English datasets. The value indicates the number of extracted samples. For example, we extract 100 sentences and 30 paragraphs from the TLDR dataset.

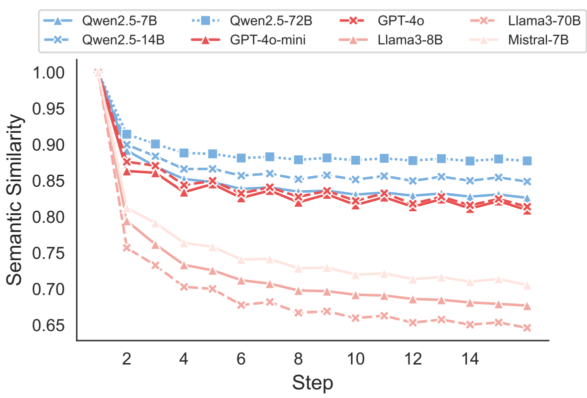

Figure 11: Similarity changes during successive paraphrasing. Qwen2.5-72B is the best at preserving meaning, while all other LLMs experience slight degradation in similarity, except during the first paraphrasing step.

perform these tasks on our paragraph dataset, calculate the textual difference of the paraphrase at each iteration with the initial text, and plot the results in Figure 12. As the number of paraphrasing steps increases, the difference between T_i and T_{i-2} decreases. After 7 steps, there is little difference between T_i and T_{i-2}.

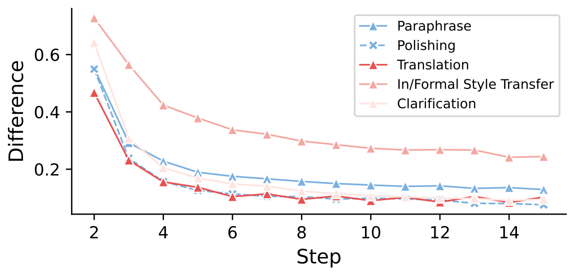

Figure 12: The trend in normalized edit distance between T_i and T_{i-2} across various tasks during the repetition process using GPT-4o-mini.

C.2 Model and Prompt variation

We continue to modify the models and prompts during paraphrasing. The chosen model set includes GPT-4o-mini, GPT-4o, Qwen2.5-7B, and Llama3-8B. Four variations of the paraphrasing prompts are provided in Table 8.

Pol.	Please polish the following text: {text}
Clar.	Please rewrite the following text in a way that is simpler and easier to understand, using clear language and shorter sentences without losing the original meaning: {text}
I/F.	Transform the following text into an informal style: {text} / Rewrite the following text in a formal style: {text}
Trans.	Please translate the following English text into Chinese: {text} / Please translate the following English text into Chinese: {text}

Table 7: Four types of prompts for extension tasks. The last two tasks involve switching between different languages and styles, separated by a semicolon.

A:	Please paraphrase the following text: {text}
B:	Please rephrase the text below: {text}
C:	Please rewrite the following text: {text}
D:	Please polish the text below: {text}

Table 8: Four variations of paraphrasing prompts. In the prompt variation experiments, a prompt is randomly selected at each step to perform the paraphrasing.

C.3 Experiment with A Complex Prompt

We conduct experiments on our paragraph-level dataset using a more complex prompt, as presented in Table 9. The resulting difference confusion matrix is shown in Figure 13.

C.4 Increasing Randomness

We measure the impact of increasing randomness on periodicity by adjusting the generation temperature. We select four temperature values: 0.6, 0.9, 1.2, and 1.5. The results are shown in Figure 14. Although the temperature increases to a

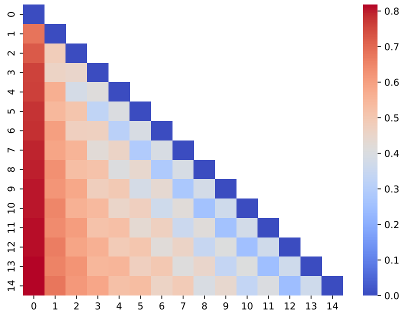

Figure 13: Difference confusion matrix for the complex prompt.

Please rewrite the following paragraph with the goal of enhancing lexical and syntactical variety without changing the original meaning. Pay attention to employing diverse vocabulary, increasing the complexity and variation of sentence structures, using different conjunctions and clause constructions to make the expression more diverse and rich, while maintaining the core information and logical coherence of the original text. Specifically, avoid repetitive sentence patterns and try to express the same ideas in different ways.

Table 9: A more complex prompt enhancing lexical and syntactical variety.

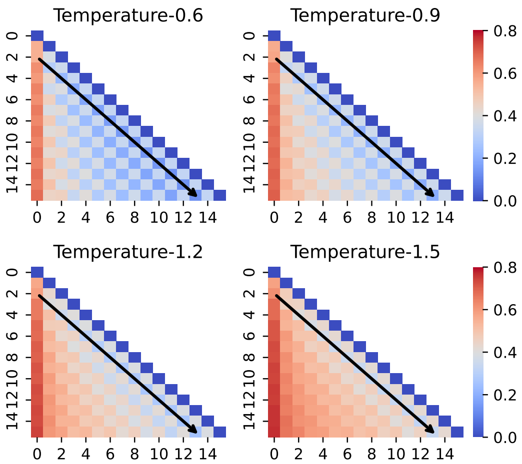

Figure 14: The difference confusion matrix for successive paraphrasing at different temperature settings, conducted by GPT-4o-mini.

very high level, the 2-periodicity still persists. Further increasing temperature will cause nonsense responses.

C.5 Sample Selection Strategies

We propose three strategies for successive paraphrasing and evaluate them across different LLMs. To assess the impact of these strategies on meaning preservation, we measure the similarity between the paraphrases T_i and their corresponding original

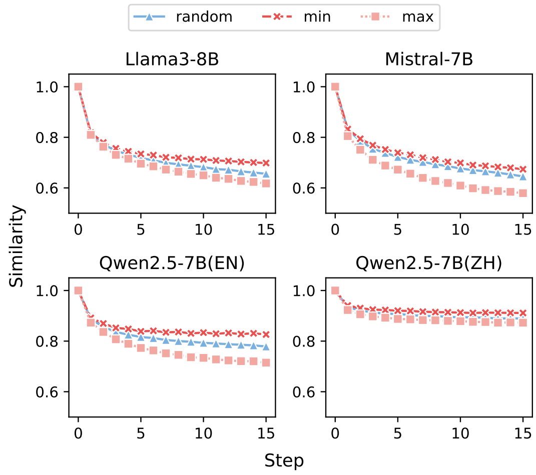

Figure 15: The similarity between paraphrases and the original texts increases during the paraphrasing process.

texts T_0 and demonstrate the result in Figure 15. By combining Figure 15 and Figure 9, we suggest that the random strategy preserves meaning significantly better than the max strategy, while also effectively alleviating periodicity.

D Case Study

We present part of the first case from our paragraph dataset for successive paraphrasing in Figure 10.